Day 2: DHCP ConfigurationTheory

Task 1 — Configure a DHCP Server in Cisco Packet Tracer

Network Setup

- Place one Server, one Switch, and two generic PCs in Cisco Packet Tracer.
- Connect Server and both PCs to the switch using copper straight-through cables.
- Use the network 192.168.10.0/24.
- Configure the DHCP server with a static address of 192.168.10.2 and subnet mask 255.255.255.0.
- Create a DHCP pool starting at 192.168.10.10 with enough addresses for the clients.

DHCP Server Configuration

- Open Server → Services → DHCP and turn DHCP service ON.
- Default Gateway: 192.168.10.1
- DNS Server: 8.8.8.8
- Start IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Maximum Number of Users: 2

Client Configuration

- On PC1, open Desktop → IP Configuration and select DHCP.
- PC1 receives: 192.168.10.10 / 255.255.255.0.
- On PC2, select DHCP.
- PC2 receives: 192.168.10.11 / 255.255.255.0.

Verification

- The two PCs receive different addresses from the DHCP pool.
- PC1 and PC2 should be able to ping each other on the same subnet.
- This verifies that the DHCP server is automatically assigning valid IP configuration.

Completed Result

- PC1 IP: 192.168.10.10
- PC2 IP: 192.168.10.11
- DHCP Server IP: 192.168.10.2
- Default Gateway: 192.168.10.1

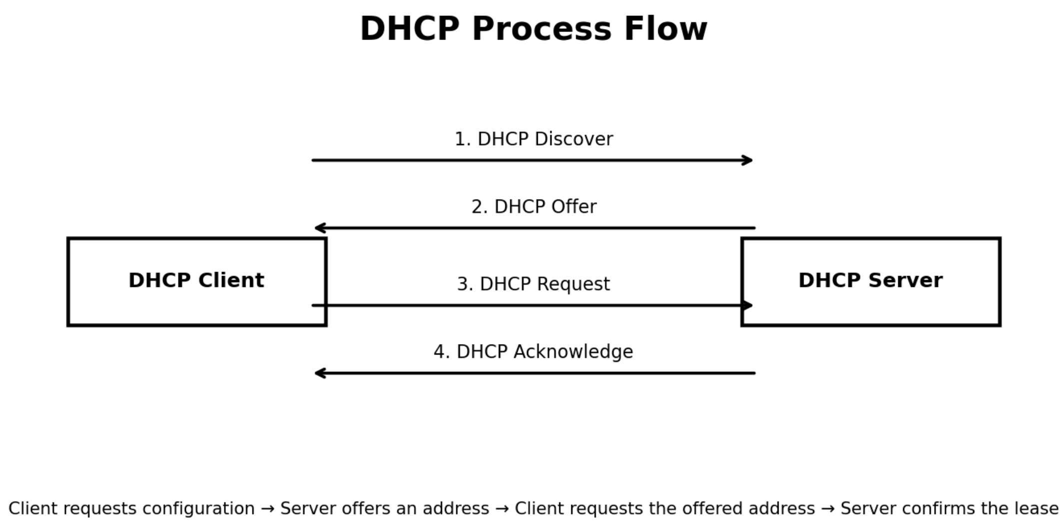
Task 2 — DHCP Process Sequence Diagram

Completed DHCP Process

- 1. Discover — The DHCP client broadcasts a DHCP Discover message to find an available DHCP server.
- 2. Offer — The DHCP server responds with a DHCP Offer containing an available IP address and network configuration.
- 3. Request — The client sends a DHCP Request indicating that it wants to use the offered address.
- 4. Acknowledge — The DHCP server sends a DHCP Acknowledgement confirming the address lease and configuration.

Sender and Receiver

- Discover: DHCP Client → DHCP Server
- Offer: DHCP Server → DHCP Client
- Request: DHCP Client → DHCP Server
- Acknowledge: DHCP Server → DHCP Client



Task 3 — DHCP Pool with Only One Available IP

Modified DHCP Pool

- Network: 192.168.20.0/24
- DHCP Server: 192.168.20.2
- Default Gateway: 192.168.20.1
- Only available DHCP address: 192.168.20.10
- Maximum number of DHCP users: 1

Client Results

- PC1 requests an address and receives 192.168.20.10.
- PC2 requests an address after the only DHCP address has already been leased.
- PC2 cannot receive another address from the one-address DHCP pool and therefore remains without a DHCP-assigned IPv4 address until an address becomes available.

Why Only One PC Succeeds

- DHCP can lease only the addresses included in the configured pool. Since the pool contains one usable address, only one client can obtain an address at a time.

Task 4 — Zomato-Style Restaurant Wi-Fi DHCP Scenario

Scenario Setup

- Create a simple Cisco Packet Tracer network with one DHCP server, one switch, one generic PC representing a laptop, and one generic PC representing a smartphone.
- Use the network 192.168.30.0/24.

DHCP Configuration

- DHCP Server: 192.168.30.2
- Default Gateway: 192.168.30.1
- DNS Server: 8.8.8.8
- DHCP Start IP: 192.168.30.10
- Subnet Mask: 255.255.255.0
- Maximum Users: 2

Assigned IP Addresses

- Restaurant Laptop: 192.168.30.10
- Restaurant Smartphone: 192.168.30.11

Why DHCP Is Useful

- Restaurant customers and staff can connect devices without manually entering IP addresses.
- DHCP automatically provides an available IP address, subnet mask, gateway and DNS information.
- It makes managing many temporary Wi-Fi clients easier and reduces configuration errors.

Conclusion

- In a Zomato-style restaurant Wi-Fi environment, DHCP allows laptops and smartphones to join the network quickly and receive valid network settings automatically.